MUNICIPALITY OF LUISIANA, LAGUNA • SYSTEM FEATURES

Clean, Current Feature List of LUISIANA-INFATRACK

Official Production Specification • Digital Twin & Municipal Governance

System: LUISIANA-INFATRACK
Document: Production Features
Version: v2.0 Production
Date: September 2026

1. Interactive 3D Cesium Map & Local GIS

SPATIAL ENGINE

- **CesiumJS 3D Globe:** Scoped to the Municipality of Luisiana with high-resolution terrain elevation (`MapTiler / Terrarium / Stadia` basemaps).
- **Road Network & Boundary Layers:** Full OpenStreetMap road networks (`luisiana-roads.geojson` , `osm-roads.ts`) and barangay administrative boundary polygons.
- **Official Geohazard Overlays:** Real-world MGB / Project NOAH hazard raster maps (`EIL 2014 & GSH 2014`) for earthquake-induced landslides, ground shaking, and flood susceptibility.
- **360° Ground-Level Panorama Viewer:** Panoramic site photo inspection (`PanoramaViewer.tsx`).
- **Offline Globe Support:** Service worker caching of terrain and map tiles for low-connectivity field operations (`globe-offline.ts`).

2. Citizen Permit & Public Services Portal

PUBLIC SERVICES

- **Citizen Portal Gateway:** Public municipal service portal (`CitizenPortal.tsx`).
- **Online Building Permit & Locational Clearance Form (`NewApplicationPage.tsx`):**
  - **Interactive parcel pinpointing** (`PinpointMapPicker.tsx`) to click and drop a pin on the lot.
  - **Document attachments** (TCT, Tax Declarations, plans, bill of materials).
  - **Draft autosave** to browser localStorage (`application-draft.ts`).
  - **Digital terms and privacy consent** with SHA-256 cryptographic logging (`ApplicationTermsModal.tsx`).
- **Real-Time Applicant Tracking Portal (`ApplicantTrackingPage.tsx`):**
  - Live milestone timeline from Zoning → Assessor → Treasury → Engineering → Fire/OBO.
  - Deficiency resolution with document re-upload capability.
  - Official Receipt (O.R.) payment submission directly from the applicant.
  - Printable clearances and official application receipts.
- **Transparency Dashboard (`TransparencyDashboard.tsx`):** Public transparency on municipal project funds, contractors, and progress.

3. Inter-Agency Municipal Governance Workflow

INTERNAL OPERATIONS

- **Zoning & Locational Clearance Office (`ZoningPermitsPage.tsx`):**
  - CLUP zoning classification verification (`clup-zones.ts`).
  - Split-screen / overlay spatial verification checking the applicant's parcel against official hazard overlays (`ZoningReviewMapView.tsx`).
  - Formal Zoning Locational Clearance issuance.
- **Municipal Assessor's Office (`AssessorOfficePage.tsx`):**
  - Real property title and tax declaration verification.
  - Automated fee schedule computation (zoning clearance, filing fee, title verification) and printable Order of Payment issuance.
- **Municipal Treasury Office (`TreasuryOfficePage.tsx`):**
  - Cashier desk with Express Teller barcode/tracking number quick lookup.
  - Payment reconciliation, Official Receipt (O.R.) logging, and End-of-Day (EOD) financial collections reporting.
- **Municipal Engineering Office (`EngineerApplicationsPage.tsx` , `ProjectMonitoringPanel.tsx`):**
  - Technical engineering plan evaluation and clearances.
  - Municipal infrastructure tracking across lifecycle phases (Planning to Maintenance).
  - Field inspection task queue (`InspectionTaskQueue.tsx`) and mobile update modal (`InspectionUpdateModal.tsx`) with offline photo upload syncing.
- **Municipal Planning & Development Coordinator (MPDC) (`PlanningPage.tsx`):**
  - Project pipeline and Annual Investment Program (AIP/LDIP) integration.
  - Multi-department committee reviews, calendar scheduling, and automated infrastructure needs assessment scoring (`planning-insights.ts`).
- **Private Engineer & Architect Portal (`PrivateEngineerPortal.tsx`):** Direct submission workspace for licensed private practitioners to upload blueprints, structural computations, and construction progress reports.
- **Barangay Officials Portal:** Localized view for barangay captains and councilors to submit grassroots project proposals and monitor works in their barangay.

4. Security, Administration & System Health

GOVERNANCE & DATA

- **10 User Roles with Role-Based Access Control (RBAC) (`Landing.tsx`):** Super Admin, MPDC, Engineer, Agriculture, Treasury Office, Negosyo Center, Barangay Official, Private Engineer, Assessor, and Public Viewer.
- **Super Admin User Management (`UserManagementPage.tsx`):**
  - Account provisioning, suspensions, credential resets, forced password resets on initial login (`ForcedPasswordResetModal.tsx`), and active online user monitoring.
- **Dynamic System Masterlist (`systemConfig.js`):** In-app configuration of fee schedules, project categories, and departments without redeploying code.
- **SLA & Citizen's Charter Monitoring (`slaMonitoring.js`):** Real-time turnaround time tracking per office to comply with Republic Act 11032 (Ease of Doing Business).
- **Tamper-Evident Audit Trail (`AuditPage.tsx`):** Cryptographically hashed audit logs tracking actor, role, IP, and timestamp.
- **Automated Backups & Cloud Sync (`backup.js` , `firebaseBackup.ts`):** Scheduled and on-demand database snapshots with Firebase Cloud Storage sync.
- **In-App Local AI Municipal Chatbot (`ProjectChat.tsx`):** Local Ollama / Llama 3.2 assistant conditioned to help staff navigate Luisiana municipal records and features.
- **PostgreSQL Database Engine (`db.js`):** Relational database schema with automated migrations and JSON file fallback.